

Bridging Composition and Transfer Operators with Locally Linear and Diffusive Kernels

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Abstract

Across non-linear dynamics, machine learning, and statistical physics, two operator viewpoints are prevalent: the Koopman perspective, which treats dynamics as deterministic composition on observables, and the Perron–Frobenius (transfer) perspective, which treats evolution as the spreading of densities. Each excel while leaving an interpretive gap between ‘pure chaos’ and ‘pure noise’.

We propose a bridge: a data-driven, two-dialed operator family. One dial controls how local the estimated drift is; the other controls how strongly residual variability is interpreted as diffusion. Sweeping the dials traces a continuous surface whose edges recover the Koopman and Perron–Frobenius pictures and whose interior supplies every intermediate mixture.

Framing this surface as a homotopy in operator space clarifies where familiar techniques—delay embeddings, diffusion maps, local-linear forecasting, stochastic model reduction—sit inside the same conceptual square. Practical diagnostics based on residual spectra reveal the minimal embedding dimension and quantify when unresolved curvature is better modeled as stochastic forcing. A continuous-time limit connects the Rose construction to classical Itô SDEs and their Fokker–Planck and Kolmogorov equations, providing a link between the data scientific and stochastic analysis perspectives.

Our goal is not to champion yet another single algorithm or approach, but provide a common scaffold. The framework helps practitioners express where they stand on the spectrum of operator models and how to move, continuously and transparently, to the representation that best fits their data and scientific questions.

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1 Local Linearizations

44 Consider the Cauchy problem

$$\begin{aligned} \frac{du}{dt} + A(u(t)) &= 0, \quad t > 0, \\ u(0) &= u_0, \quad u_0 \in U \subset X, \end{aligned} \tag{1}$$

46 where A is an operator (possibly nonlinear and state-dependent) that generates a contraction semi-group $T(h)$ on the Banach space $\mathcal{X}$.

48 In many practical settings, one only has access to discrete observations of the state. Let these observations be arranged in the data matrix

$$X_t = \begin{bmatrix} x(t) \\ x(t-h) \\ x(t-2h) \\ \vdots \end{bmatrix}, \tag{2}$$

where the rows correspond to states sampled at time intervals of length h .

50 A first approach is to attempt a global linearization via a least-squares solution to,

$$X_t \approx X_{t-h} T(h), \tag{3}$$

$$\text{or equivalently, } u(t) \approx T(h) x(t-h). \tag{4}$$

52 If the residuals from this approximation are well within an acceptable and prescribed tolerance—then it is reasonable to assume that the dynamics are well represented by a linear operator. In such a

case, one can recover the global generator via

$$A = \lim_{h \rightarrow 0^+} \frac{1}{h} (T(h) - I). \quad (5)$$

54 This approach is particularly effective when A is linear or nearly linear. The contraction property
aids in establishing the existence and uniqueness of solutions, however it also means that a global
56 linearization of A may be unattainable when its dependence on $u(t)$ is significant with residuals
remaining unacceptably large.

58 A global linearization may fail to capture the system's behavior when A is state dependent. In such
case a local linearization may be preferable.

60 To construct a local approximation we can introduce a weighting operator W designed to select
or emphasize those rows in X_t that are dynamically relevant. The idea is to filter the data so
62 that regions exhibiting dynamics similar to those surrounding $x(t)$ are highlighted, typically by
considering observations local to $x(t)$ under some metrization of $\mathcal{X}$.

64 If W is constructed using a localization kernel, $k_\theta(x(j), x(i))$, where θ parametrizes what "local"
means, then the weighted observations are,

$$W_\theta X_t, \quad (6)$$

66 and the local approximation is expressed as

$$W_\theta X_t \approx W_\theta X_{t-h} \hat{T}(h, \theta, t), \quad (7)$$

where $\hat{T}(h, \theta, t)$ is the locally linearized evolution operator valid in a neighborhood of $x(t)$. If a
68 weighted pseudoinverse is employed,

$$A(x(t), \theta, h) = \left((W_\theta X_{t-h})^\dagger (W_\theta X_t) - I \right) \quad (8)$$

where $(W X_{t-h})^+$ denotes the pseudoinverse of $W X_{t-h}$. Now recovery of the generator $A(x(t))$
70 requires the spatial limit in θ , followed by the temporal limit in h ,

$$A(x(t)) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\lim_{\theta \rightarrow 0} \hat{T}(h, \theta, t) - I \right) \quad (9)$$

If T is (i) deterministic (i.e. $\theta \rightarrow 0$ iff, in the weak sense, $k_\theta \rightarrow \delta$), and (ii) locally linear (i.e.
72 $T = \sum_{k=0}^{\infty} \frac{h^k}{k!} A(h, \theta, x_j)^k$ converges where required), then this is a nice framework for approxim-
ing the state-dependent generator $A(x(t))$ in the presence of features precluding a global linear
74 model.

However, in the absence of any strong *a priori* intuition regarding even the most general assumptions
76 of the generating process, most notably stochasticity and the rate of decay of higher order terms,
deciding how to approach these limits (and thus what constitutes a *good* linearization) is not
78 easy.

2 Composition and Transfer Operators

Definition 2.1 (Composition (Koopman) operator). Let $T_h : X \rightarrow X$ be the time- h map of the flow (1). For any square-integrable observable $f \in L^2(X, \mu)$ define

$$(Kf)(x) = f(T_h x)$$

- K is linear but acts on an *infinite-dimensional* function space even when $\dim X < \infty$.
- If T_h preserves μ then K is unitary and its spectrum lies on the unit circle.
- No smoothing occurs: higher-order nonlinearities of the flow are implicit in composition, so K captures sharp deterministic features but is blind to stochasticity.
- For the Dirac kernel $p_\delta(x' | x) = \delta(x' - T_h x)$, K is the *left* adjoint of the deterministic Perron-Frobenius operator P_δ (see (10) below).

Global linearisation (3) corresponds to seeking a finite-rank approximation of K .

Definition 2.2 (Transfer (Perron-Frobenius) operator). Let $p(x' | x)$ be a Markov transition density. For any $\rho \in L^1(X, \mu)$ set

$$(P\rho)(x') = \int_X \rho(x) p(x' | x) d\mu(x) \quad (10)$$

- P is positive and mass-preserving: $\int (P\rho) d\mu = \int \rho d\mu$.
- Under mild regularity P is a strict contraction on the mean-zero subspace $L_0^2(X)$ whenever p has non-zero width; its spectral radius is then < 1 .
- A non-trivial width encodes diffusion, measurement error, or model misspecification. In local linearisation the residual cloud $r_t^{(\theta)}$ naturally induces such a kernel.
- For the deterministic kernel $p_\delta(x' | x) = \delta(x' - T_h x)$, P_δ is the *right* adjoint of the Koopman operator (2.1).

Throughout we fix a reference measure μ ; all L^p spaces, inner products, and adjoints are taken with respect to μ .

Two extreme lenses. Koopman K captures deterministic curvature but ignores noise, whereas a non-delta transfer operator P absorbs all residual variation as diffusion. Global least-squares (3) implicitly demands that K admit an accurate finite-rank approximation; large residuals signal failure. Conversely, shunting all residual energy into P treats it as pure diffusion.

Need for a continuous bridge. We therefore seek a *continuous family* of operators

$$(\theta, \sigma) \mapsto (\mathcal{R}_{\theta, \sigma}^b, \mathcal{R}_{\theta, \sigma}^\sharp)$$

such that $\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \mathcal{R}_{\theta, \sigma}^b = K$ and $\lim_{\theta, \sigma \rightarrow \infty} \mathcal{R}_{\theta, \sigma}^\sharp = P_\infty$ (the maximally diffusive rank-one PF operator). The construction is *local* (uses weighted neighbourhoods) and *data-adaptive* (learns residual covariance), producing a two-parameter homotopy surface whose θ -axis refines deterministic drift and whose σ -axis injects diffusion.

The next section formalises this via the push-forward / pull-back pair $(\mathcal{R}_{\theta,\sigma}^\sharp, \mathcal{R}_{\theta,\sigma}^\flat)$ introduced in Definition (3.1).
110

3 From Koopman and PF to a Two-Dial Family

The preceding section set the deterministic pull-back (Koopman) and stochastic push-forward (Perron–Frobenius) operators at opposite ends of the modeling spectrum. We now construct a
112 *two-parameter* family of operators
114

$$(\theta, \sigma) \mapsto (\mathcal{R}_{\theta,\sigma}^\flat, \mathcal{R}_{\theta,\sigma}^\sharp), \quad (\theta, \sigma) \in (0, \infty) \times (0, \infty)^1 \quad (11)$$

with the following properties:

- 116 1. $\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \mathcal{R}_{\theta,\sigma}^\flat = K$ and, dually, $\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \mathcal{R}_{\theta,\sigma}^\sharp = P_\delta$ (the deterministic PF operator with Dirac kernel).
- 118 2. $\lim_{\theta, \sigma \rightarrow \infty} \mathcal{R}_{\theta,\sigma}^\sharp = P_\infty$ where P_∞ is the rank-one transfer operator whose kernel is a spatially uniform density, and $\mathcal{R}_{\theta,\sigma}^\flat$ collapses observables to the corresponding constant.
- 120 3. The map $(\theta, \sigma) \mapsto (\mathcal{R}_{\theta,\sigma}^\flat, \mathcal{R}_{\theta,\sigma}^\sharp)$ is jointly continuous in the L^2 -operator norm; varying θ (locality) and σ (residual blur) in either order yields the same result, giving a *homotopy of homotopies*—a filled square in the 2-category $\text{End}(L^2(X, \mu))$.
122

We call this two-dial construction the **Rose operator family**. Operationally it couples

- 124 • a *locally weighted linear predictor* $C_x^{(\theta)}$ that refines deterministic drift as $\theta \downarrow 0$, and
- an *empirical residual kernel* governed by g_σ that injects increasing diffusion as $\sigma \uparrow \infty$.

126 Because the θ - and σ -axes commute, one may independently *sharpen drift* (decrease θ) or *blur noise* (increase σ), tracing paths across the square that suit the data at hand. In the interior of
128 this “Rose surface” lie operator pairs that blend deterministic curvature with state-dependent or homogeneous diffusion—precisely the regime where, presumably, most representations arise.

3.1 Two-parameter Rose operator $\mathcal{R}_{\theta,\sigma}$

Locality kernel ϕ (state space). Fix a radial profile $\phi : [0, \infty) \rightarrow \mathbb{R}_+$ satisfying

$$\int_{\mathbb{R}^d} \phi(\|u\|) du = 1 \quad \phi(r) = \phi(-r) \quad \int r^2 \phi(r) dr < \infty \quad \|\phi_{\lambda_1} - \phi_{\lambda_2}\|_{L^1} \leq L |\lambda_1 - \lambda_2| \quad (12)$$

132 with $\phi_\lambda(r) := \lambda^{-d} \phi(r/\lambda)$. Typical choices include Gaussian, Epanechnikov, tri-weight, or a heat kernel on a Riemannian manifold. For bandwidth $\theta > 0$ define the neighbourhood weight

$$w_\theta(\|x_t - x\|) = \phi_\theta(\|x_t - x\|) \quad (13)$$

¹Because $(0, \infty)$ is homeomorphic to $[0, 1)$ via $u \mapsto \frac{1}{(1+u)}$ the homotopy can be re-parameterised to the unit interval without affecting continuity; we retain the positive bandwidth scales for engineering clarity.

134 **Local drift and residuals.** With $W_\theta = \text{diag}(w_\theta(\|x_t - x\|))$ and $X = [x_0, \dots, x_{N-1}]$, $Y = [x_h, \dots, x_{N-1+h}]$, set

$$C_x^{(\theta)} = Y_\theta X_\theta^\top (X_\theta X_\theta^\top)^\dagger, \quad r_t^{(\theta)}(x) = x_{t+h} - C_x^{(\theta)} x_t \quad (14)$$

136 **Residual kernel g_σ (noise space).** Let $g_\sigma : \mathbb{R}^d \rightarrow \mathbb{R}_+$ be radial with $\int g_\sigma = 1$ and $g_\sigma \rightarrow \delta$ as $\sigma \downarrow 0$ (e.g. Gaussian, Huber, Tukey). Define first

$$Z_{\theta,\sigma}(x) = \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}(x)) \quad (15)$$

138 and then the weighted residual mean

$$\bar{r}_x^{(\theta,\sigma)} = \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}(x)) r_t^{(\theta)} \quad (16)$$

as well as the covariance

$$\Sigma_x^{(\theta,\sigma)} = \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|) g_\sigma(r_t^{(\theta)}(x)) (r_t^{(\theta)} - \bar{r}_x^{(\theta,\sigma)})(r_t^{(\theta)} - \bar{r}_x^{(\theta,\sigma)})^\top \quad (17)$$

Transition kernel.

$$k_{\theta,\sigma}(y \mid x) = \mathcal{N}(y; C_x^{(\theta)} x + \bar{r}_x^{(\theta,\sigma)}, \Sigma_x^{(\theta,\sigma)}) \quad (18)$$

140 **Composition and Transfer Adjunctions.**

Definition 3.1 (Push-forward and pull-back Rose operators). For the kernel $k_{\theta,\sigma}$ defined above
142 set,

$$(\mathcal{R}_{\theta,\sigma}^\# \rho)(y) := \int_X k_{\theta,\sigma}(y \mid x) \rho(x) d\mu(x), \quad (\mathcal{R}_{\theta,\sigma}^\flat f)(x) := \int_X f(y) k_{\theta,\sigma}(y \mid x) d\mu(y) \quad (19)$$

Then $\mathcal{R}_{\theta,\sigma}^\flat$ is the Hilbert-space adjoint of $\mathcal{R}_{\theta,\sigma}^\#$ with respect to the $L^2(X, \mu)$ pairing,

$$\langle f, \mathcal{R}_{\theta,\sigma}^\# \rho \rangle_{L^2} = \langle \mathcal{R}_{\theta,\sigma}^\flat f, \rho \rangle_{L^2} \quad \text{for all } f, \rho \in L^2(X, \mu) \quad (20)$$

144 3.2 Homotopy theorems for the two-dial Rose family

The results below make precise the idea that the two-parameter Rose operators

$$(\theta, \sigma) \mapsto (\mathcal{R}_{\theta,\sigma}^\flat, \mathcal{R}_{\theta,\sigma}^\#) \quad (21)$$

146 form a *continuous square* whose left-hand edge collapses to the Koopman / deterministic-PF pair and whose right-hand edge collapses to the rank-one, maximally diffusive PF pair.

148 **Assumption 3.1** (Standing hypotheses).

1. $X \subset \mathbb{R}^d$ is a compact C^∞ manifold equipped with a finite Borel measure μ .

- 150 2. $T_h : X \rightarrow X$ is the time- h map of a C^2 flow with bounded first and second derivatives,
 $\sup_x \|DT_h(x)\| \leq M_1$, $\sup_x \|D^2T_h(x)\| \leq M_2$.
- 152 3. The locality profile ϕ is radial, integrable, symmetric, finite-moment, and Lipschitz in scale
 (Section 3.1).
- 154 4. The residual-weight profile g_σ is radial with $\int g_\sigma = 1$ and $g_\sigma \rightarrow \delta$ in L^1 as $\sigma \downarrow 0$.

Theorem 3.1 (Two-parameter Rose homotopy). *Under Assumption 3.1, in the population (infinite-data) setting, the following statements hold.*

(i) **Local bias and covariance bounds**

158 *For every $x \in X$ and $\sigma > 0$*

$$\|C_x^{(\theta)} - DT_h(x)\| \leq c_1 \theta, \quad \|\Sigma_x^{(\theta, \sigma)}\| \leq c_2 (\theta^2 + \sigma^2)^2 \quad (22)$$

(ii) **Deterministic corner**

$$\lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \|\mathcal{R}_{\theta, \sigma}^\flat - K\|_{L^2 \rightarrow L^2} = 0, \quad \lim_{\theta \rightarrow 0, \sigma \rightarrow 0} \|\mathcal{R}_{\theta, \sigma}^\sharp - P_\delta\|_{L^2 \rightarrow L^2} = 0 \quad (23)$$

160 (iii) **Maximally diffusive edge**

$$\lim_{\theta, \sigma \rightarrow \infty} \|\mathcal{R}_{\theta, \sigma}^\sharp - P_\infty\|_{L^2 \rightarrow L^2} = 0, \quad P_\infty \rho = \left(\int_X \rho d\mu \right) p_\infty \quad (24)$$

where p_∞ is the state-independent limit density.

162 (iv) **Separate Lipschitz continuity**

There exist constants $C_\theta, C_\sigma > 0$ such that

$$\|\mathcal{R}_{\theta_1, \sigma}^\flat - \mathcal{R}_{\theta_2, \sigma}^\flat\|_{L^2 \rightarrow L^2} \leq C_\theta |\theta_1 - \theta_2|, \quad \|\mathcal{R}_{\theta, \sigma_1}^\flat - \mathcal{R}_{\theta, \sigma_2}^\flat\|_{L^2 \rightarrow L^2} \leq C_\sigma \|g_{\sigma_1} - g_{\sigma_2}\|_{L^1} \quad (25)$$

164 and the same bounds hold for the push-forward operators $\mathcal{R}^\sharp$.

(v) **Commutativity (2-cell)**

166 For any $\theta_2 \geq \theta_1 > 0$ and $\sigma_2, \sigma_1 > 0$,

$$\mathcal{R}_{\theta_2, \sigma_2} = \mathcal{R}_{\theta_2, \sigma_2 - \sigma_1} \circ \mathcal{R}_{\theta_1, \sigma_1} = \mathcal{R}_{\theta_1, \sigma_1} \circ \mathcal{R}_{\theta_2 - \theta_1, \sigma_2} \quad (26)$$

168 whenever g_σ is Gaussian; more generally the two one-parameter homotopies commute up to a vanishing L^2 -operator error of order $O(|\theta_2 - \theta_1| \|g_{\sigma_2} - g_{\sigma_1}\|_{L^1})$.

170 Constants $c_1, c_2, C_\theta, C_\sigma$ depend only on M_1, M_2 , the second moment of ϕ , and integrability bounds for g .

172 *Sketch of proof.* (i) A second-order Taylor expansion plus symmetry of ϕ yields the bias bound; the covariance bound follows by inserting $g_\sigma \leq \|g_\sigma\|_\infty$ and taking second moments.

- (ii) As $\theta, \sigma \downarrow 0$, $C_x^{(\theta)} \rightarrow DT_h(x)$ and $\Sigma_x^{(\theta, \sigma)} \rightarrow 0$; the Gaussian TV estimate gives $k_{\theta, \sigma} \rightarrow \delta_{T_h x}$ in L^1 ,
 174 hence operator convergence.
- (iii) With $\theta, \sigma \uparrow \infty$, locality weights flatten and g_σ acts as heavy blur: the kernel columns converge
 176 to the state-independent density p_∞ .
- (iv) Lipschitz in θ is inherited from Lipschitz scale dependence of ϕ ; Lipschitz in σ follows from
 178 $\|g_{\sigma_1} - g_{\sigma_2}\|_{L^1}$.
- (v) For Gaussian g_σ , convolution is additive in variances, so ordering of the two dials commutes exactly; for general radial kernels one obtains the stated error estimate by Young's inequality. $\square$
 180

$$\begin{array}{ccc}
 (K, P_\delta) & \xrightarrow{\sigma} & (R_{\theta, \infty}^b, R_{\theta, \infty}^\#) \\
 \downarrow \theta & \boxed{H(\theta, \sigma)} & \downarrow \theta \\
 (R_{\infty, \sigma}^b, R_{\infty, \sigma}^\#) & \xrightarrow{\sigma} & (P_\infty^*, P_\infty)
 \end{array}$$

Figure 1: Homotopy square for 2-cell (homotopy of homotopies) that continuously deforms the Koopman operator K into the maximally diffusive transfer operator P_∞ .

4 Continuous-Time Limit and Stochastic Interpretation

182 The discrete-time Rose operators $(\mathcal{R}_{\theta, \sigma}^b, \mathcal{R}_{\theta, \sigma}^\#)$ approximate, in the small-lag limit, the semigroup of an Itô diffusion. This section formalises that link and shows how the local bandwidth-weighted
 184 quantities $C_x^{(\theta)}$ and $\Sigma_x^{(\theta, \sigma)}$ furnish non-parametric estimators of the drift $a(x)$ and diffusion tensor $D(x)$ of the underlying continuous dynamics.

186 Throughout this section $h > 0$ denotes the sampling interval and we explicitly write $C_x^{(\theta, h)}$, $r_t^{(\theta, h)}$, $\Sigma_x^{(\theta, \sigma, h)}$ to highlight the dependence on h . Assumptions 3.1 remain in force.

188 **Lemma 4.1** (First-order expansion). *For every fixed $\theta > 0, \sigma > 0$ and anchor $x \in X$*

$$C_x^{(\theta, h)} = I + a(x)h + O(h\theta + h^2), \quad \Sigma_x^{(\theta, \sigma, h)} = 2D(x)h + O(h\theta^2 + h\sigma^2) \quad (27)$$

190 where $a(x) = \dot{X}_t|_{t=0}$ is the deterministic drift and $D(x) = \frac{1}{2}B(x)B(x)^\top$ is the diffusion tensor of the Itô SDE $dX_t = a(X_t)dt + B(X_t)dW_t$.

192 *Sketch.* Second-order Taylor expansion of T_h yields $T_h x_t = x + a(x)h + O(h\|x_t - x\| + h^2)$. Weighted first-moment symmetry of ϕ eliminates the linear $x_t - x$ term, giving the stated drift expansion.

For the covariance, subtract $\bar{r}_x^{(\theta, \sigma, h)}$ and take second moments; the leading term is $2D(x)h$ and the weights introduce the stated $O(h\theta^2 + h\sigma^2)$ correction. $\square$

Theorem 4.1 (Discrete-to-continuous limit). *Fix $\theta, \sigma > 0$ and let $h \downarrow 0$. For any $f \in C^2(X)$ and $\rho \in C^2(X)$,*

$$\frac{\mathcal{R}_{\theta, \sigma}^{b, h} f - f}{h} \longrightarrow \mathcal{L}^b f, \quad \frac{\mathcal{R}_{\theta, \sigma}^{\sharp, h} \rho - \rho}{h} \longrightarrow \mathcal{L}^\sharp \rho \quad \text{in } L^2(X, \mu) \quad (28)$$

where the Kolmogorov generators are

$$\mathcal{L}^b f = a \cdot \nabla f + D : \nabla \nabla f, \quad \mathcal{L}^\sharp \rho = -\nabla \cdot (a \rho) + \nabla \cdot (D \nabla \rho) \quad (29)$$

corresponding to the Itô SDE above.

Sketch. Expand $f(y)$ to second order around x and integrate against the Gaussian kernel with mean $x + a(x)h$ and variance $2D(x)h$; the first-order terms reproduce $\mathcal{L}^b f$. An identical calculation with adjoint pairing yields the forward generator $\mathcal{L}^\sharp$. $\square$

Remark 4.1. Lemma 4.1 yields *data-driven, bandwidth-controlled* estimators for both coefficients of the limiting SDE, namely $\hat{a}(x) = (C_x^{(\theta, h)} - I)/h$ for the drift and $\hat{D}(x) = \frac{1}{2} \Sigma_x^{(\theta, \sigma, h)}/h$ for the diffusion tensor. Once these fields are available, the generator pair $(\mathcal{L}^b, \mathcal{L}^\sharp)$ provides two complementary routes to uncertainty propagation: forward evolution of densities through the Fokker–Planck equation and simulation of sample paths via the Itô SDE, with the discrete Rose operators furnishing time- h stepping schemes that are consistent with both perspectives.

5 Fitting

5.1 Bandwidth selection via validation likelihood

To turn the candidate grid $(\theta, \sigma) \in \Theta \times \Sigma$ in Algorithm 1 into a single “best” pair we minimise the *negative Gaussian log-likelihood* on a hold-out set $\mathcal{V} \subset \{0, \dots, N-1\}$. Throughout x denotes a fixed anchor; superscripts (θ, σ) are omitted when clear.

Predictive distribution on the validation set. With drift matrix $C_x^{(\theta)}$, residual mean $\bar{r}_x^{(\theta, \sigma)}$, and covariance $\Sigma_x^{(\theta, \sigma)}$ computed from the *training* points $t \notin \mathcal{V}$, the one-step prediction for a validation pair (x_t, x_{t+h}) is Gaussian

$$x_{t+h} \mid x_t \sim \mathcal{N}(\hat{m}_t, \Sigma_x^{(\theta, \sigma)}), \quad \hat{m}_t := C_x^{(\theta)} x_t + \bar{r}_x^{(\theta, \sigma)}.$$

Per-sample log-likelihood. Up to an additive constant $\frac{d}{2} \log 2\pi$ independent of (θ, σ) , the negative log-likelihood of observing $y = x_{t+h}$ is

$$\ell_t(\theta, \sigma) = \frac{1}{2} \left[(y - \hat{m}_t)^\top \Sigma^{-1} (y - \hat{m}_t) + \log \det \Sigma \right]$$

- The first term is *Mahalanobis error* that measures how far the prediction misses, normalized by the local covariance.
- The second term penalizes excessively broad covariances that would otherwise drive the Mahalanobis error down.

222 **Validation loss.** Summing ℓ_t over the validation indices gives the criterion

$$\mathcal{L}(x; \theta, \sigma) = \sum_{t \in \mathcal{V}} \left[(x_{t+h} - \hat{m}_t)^\top \Sigma^{-1} (x_{t+h} - \hat{m}_t) + \log \det \Sigma \right]$$

224 The optimal bandwidths for anchor x are $(\theta^*(x), \sigma^*(x)) = \arg \min_{(\theta, \sigma)} \mathcal{L}(x; \theta, \sigma)$. Aggregating the loss across anchors before minimization yields global bandwidths (θ^*, σ^*) .

Interpretation.

- 226 1. When $\theta, \sigma \downarrow 0$ in a purely deterministic system, both the Mahalanobis term and $\log \det \Sigma$ shrink, driving the loss down and reproducing the Koopman limit.
- 228 2. If irreducible noise is present, shrinking θ past the curvature scale stops improving the Mahalanobis term, while $\log \det \Sigma$ inflates; the optimum stabilizes at a positive σ , signaling diffusion.
- 230 3. The loss is proportional to the *cross-entropy* between the empirical distribution of the validation pairs and the Gaussian model induced by (θ, σ) ; thus minimizing it selects the bandwidths that best balance drift bias, residual variance and uncertainty calibration.

234 Proof sketches for the validation–loss heuristics

Throughout, write $\hat{\Sigma} := \Sigma_x^{(\theta, \sigma)}$, $\hat{m}_t := C_x^{(\theta)} x_t + \bar{r}_x^{(\theta, \sigma)}$. The validation loss for anchor x is

$$\mathcal{L}(x; \theta, \sigma) = \sum_{t \in \mathcal{V}} \left[(x_{t+h} - \hat{m}_t)^\top \hat{\Sigma}^{-1} (x_{t+h} - \hat{m}_t) + \log \det \hat{\Sigma} \right].$$

236 We take $h > 0$ fixed and consider $\theta, \sigma \downarrow 0$ (simultaneously) or $\theta \downarrow 0$ with σ fixed.

(1) **Deterministic limit** $\theta, \sigma \downarrow 0$. Under Assumption 3.1 the flow is C^2 , so Lemma 4.1 gives $\|\hat{m}_t - T_h x_t\| = O(h\theta)$ and $\|\hat{\Sigma}\| = O(h\theta^2 + h\sigma^2)$. Inserting these into $\mathcal{L}$ and using $\hat{\Sigma}^{-1} = O((h\theta^2 + h\sigma^2)^{-1})$, each Mahalanobis term is $O(h\theta^2 + h\sigma^2)$ and $\log \det \hat{\Sigma} = d \log(h\theta^2 + h\sigma^2) + O(1)$. Therefore, both contributions decrease monotonically to 0 as $\theta, \sigma \rightarrow 0$, reproducing the Koopman corner.

(2) **Irreducible noise** ($\theta \downarrow 0$, σ fixed). Assume the observations satisfy $x_{t+h} = T_h x_t + \eta_t$ with $\eta_t \sim \mathcal{N}(0, \Sigma_0)$, $\Sigma_0 \succ 0$. Then $\hat{\Sigma} = \Sigma_0 + O(h\theta^2)$ and $\hat{m}_t = T_h x_t + O(h\theta)$. Shrinking θ below the curvature scale does not change the leading term Σ_0 , so $\log \det \hat{\Sigma}$ stabilizes, while the Mahalanobis term is already $O(1)$. Decreasing σ below the intrinsic noise level inflates the Mahalanobis term (under-dispersion), whereas increasing σ inflates $\log \det \hat{\Sigma}$ (over-dispersion). Hence the loss attains its minimum at a positive $\sigma^* \approx \sqrt{\text{tr} \Sigma_0}$, signalling diffusion.

(3) **Information–theoretic interpretation.** Let $p_{\theta, \sigma}(y \mid x) = \mathcal{N}(y; \hat{m}(x), \hat{\Sigma}(x))$ denote the Rose kernel, and q_{emp} the empirical distribution of validation pairs. Then up to an additive constant

$$\frac{1}{|\mathcal{V}|} \mathcal{L}(x; \theta, \sigma) = H(q_{\text{emp}}, p_{\theta, \sigma}) = H(q_{\text{emp}}) + \text{KL}(q_{\text{emp}} \parallel p_{\theta, \sigma}),$$

250 where H is cross-entropy and KL the Kullback–Leibler divergence. Since $H(q_{\text{emp}})$ is constant in
 252 (θ, σ) , minimising $\mathcal{L}$ is equivalent to minimising $\text{KL}(q_{\text{emp}} \| p_{\theta, \sigma})$: the chosen bandwidths give the
 Gaussian model closest (in information distance) to the empirical distribution, balancing drift bias,
 residual variance, and uncertainty calibration.

254 **Robust and regularized variants.** Heavy-tailed residuals can be handled by replacing the
 Mahalanobis quadratic with a Huber distance. If spatial smoothness of (θ^*, σ^*) is desired, add a
 256 penalty $\gamma(\theta^{-p} + \sigma^{-q})$ to $\mathcal{L}$, tuning γ by outer cross-validation.

Algorithm 1: Rose–kernel fitting with bandwidth selection

Input: time-lag pairs $\{(x_t, x_{t+h})\}_{t=0}^{N-1}$;
 candidate locality radii $\Theta = \{\theta_k\}_{k=1}^K$;
 residual widths $\Sigma = \{\sigma_\ell\}_{\ell=1}^L$;
 hold-out index set $\mathcal{V}$ (validation)
Output: selected $(\theta^*(x), \sigma^*(x))$;
 $C_x^{(\theta^*)}$,
 $\Sigma_x^{(\theta^*, \sigma^*)}$,
 eigenvalues $\{\lambda_i(x)\}$

```

1  for  $x \in \text{AnchorSet}$  do
2     $\text{bestLoss} \leftarrow \infty$ ;
3    for  $\theta \in \Theta$  do
4      compute weights  $w_t \leftarrow \phi_\theta(\|x_t - x\|)$ 
5       $C \leftarrow (YW)(XW)^\dagger$ 
6       $r_t \leftarrow x_{t+h} - Cx_t$ 
7      for  $\sigma \in \Sigma$  do
8         $g_t \leftarrow g_\sigma(r_t)$ ,
9        form  $\Sigma$  and  $\bar{r}$  as in Section 3.1
10       // --- validation loss ---
11        $\text{loss} \leftarrow \sum_{t \in \mathcal{V}} \|x_{t+h} - Cx_t - \bar{r}\|_{\Sigma^{-1}}^2 + \log \det \Sigma$ 
12       if  $\text{loss} < \text{bestLoss}$  then
13         store  $\theta^* \leftarrow \theta$ ,  $\sigma^* \leftarrow \sigma$ ,  $C_x^{(\theta^*)} \leftarrow C$ ,  $\Sigma_x^{(\theta^*, \sigma^*)} \leftarrow \Sigma$ ,  $\text{bestLoss} \leftarrow \text{loss}$ 
14       end
15     end
16    $\{\lambda_i(x)\} \leftarrow \text{eigvals}(\Sigma_x^{(\theta^*, \sigma^*)})$ 
17 end

```

References

A Notation & Conventions

PX	probability measures on $(X, \mathcal{F})$
G	Giry monad (B.15)
$x \mapsto \delta_x$	Dirac embedding
$\rightsquigarrow$	kernel arrow (Def. B.13)
$\flat, \sharp$	musical conversions (Def. ??)
P_t, L	Markov semigroup & generator (B.23–B.24)
k_Δ, T	LWLS kernel & Rose operator (??–??)
Δ	discrete time step in data

B Definitions

B.1 Measure & Probability Foundations

Definition B.1 (σ -Algebra). Given a set X , a σ -algebra $\mathcal{F} \subseteq 2^X$ is a non-empty collection of subsets closed under complementation and countable unions (hence also countable intersections).

Definition B.2 (Measurable Space). A *measurable space* is a pair $(X, \mathcal{F})$ where $\mathcal{F}$ is a σ -algebra on X (Definition B.1).

Definition B.3 (Measurable Map). If $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ are measurable spaces, a map $f : X \rightarrow Y$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{G}$.

Definition B.4 (Measure & Probability Measure). Let $(X, \mathcal{F})$ be a measurable space. A *measure* is a countably additive map $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$. It is a *probability measure* if $\mu(X) = 1$. We write PX for the set of all probability measures on $(X, \mathcal{F})$.

Definition B.5 (Push-Forward Measure). Given a measurable $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $\mu \in PX$, the *push-forward* measure $f_*\mu \in PY$ is defined by $(f_*\mu)(B) = \mu(f^{-1}(B))$ for $B \in \mathcal{G}$.

Definition B.6 (Pull-Back Measure). Given a measurable $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $\mu \in PY$, the *pull-back* measure $f^*\mu \in PX$ is defined by $f^*\mu(A) := \mu(f(A))$ for $A \in \mathcal{F}$.

B.2 Category Theory

Definition B.7 (Category). A *category* $\mathcal{C}$ consists of a class of objects, hom-sets $\text{Hom}_{\mathcal{C}}(A, B)$ for each ordered pair (A, B) , identity morphisms id_A , and associative composition, satisfying the usual unit laws.

Definition B.8 (Functor). Let $\mathcal{C}, \mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns to each object A of $\mathcal{C}$ an object FA of $\mathcal{D}$ and to each morphism $f : A \rightarrow B$ a morphism $Ff : FA \rightarrow FB$, preserving identities and composition.

Definition B.9 (Finite products in a category). Let $\mathcal{C}$ be a category and $\{X_i\}_{i=1}^n$ a finite family of objects in $\mathcal{C}$ ($n \geq 0$).

- A *product* of the family is a pair

$$\left(X, \{ \pi_i \}_{i=1}^n \right), \quad \pi_i : X \longrightarrow X_i \text{ in } \mathcal{C},$$

such that for every object Y and every collection of morphisms $\{ f_i : Y \rightarrow X_i \}_{i=1}^n$ there exists a *unique* morphism

$$f : Y \longrightarrow X \quad \text{with} \quad \pi_i \circ f = f_i \quad (i = 1, \dots, n).$$

- When it exists, the product is unique up to a unique isomorphism; we write $X = \prod_{i=1}^n X_i$ and call the π_i the *projection morphisms*.
- For $n = 0$ the empty product is any *terminal object* $1 \in \mathcal{C}$ (unique up to unique isomorphism).

$$\begin{array}{ccccc} & & Y & & \\ & f_1 \swarrow & \vdots f \downarrow & \searrow f_2 & \\ X_1 & \xleftarrow{\pi_1} & X_1 \times X_2 & \xrightarrow{\pi_2} & X_2 \end{array}$$

The dashed arrow f is the *unique* morphism rendering all triangles commutative: $\pi_i \circ f = f_i$ for $i = 1, \dots, n$. This diagram visualises the universal property that characterises the product object $X = \prod_{i=1}^n X_i$.

Definition B.10 (Natural Transformation). Given functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a *natural transformation* $\eta : F \Rightarrow G$ assigns to every object A a morphism $\eta_A : FA \rightarrow GA$ such that $Gf \circ \eta_A = \eta_B \circ Ff$ for all $f : A \rightarrow B$ in $\mathcal{C}$.

Definition B.11 (Monad). A *monad* (T, η, μ) on a category $\mathcal{C}$ comprises an endofunctor $T : \mathcal{C} \rightarrow \mathcal{C}$, a unit natural transformation $\eta : \text{id}_{\mathcal{C}} \Rightarrow T$ and a multiplication $\mu : T^2 \Rightarrow T$ satisfying $\mu \circ T\eta = \mu \circ \eta T = \text{id}_T$ and $\mu \circ T\mu = \mu \circ \mu T$.

Definition B.12 (Category **Meas**). Objects are measurable spaces; morphisms are measurable maps (Definition B.3).

Definition B.13 (Markov Kernel). Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A *Markov kernel* $\kappa : X \rightsquigarrow Y$ is a map $\kappa : X \times \mathcal{G} \rightarrow [0, 1]$ such that

1. for each $x \in X$, $B \mapsto \kappa(B \mid x)$ is a probability measure on $(Y, \mathcal{G})$;
2. for each $B \in \mathcal{G}$, $x \mapsto \kappa(B \mid x)$ is $\mathcal{F}$ -measurable.

Definition B.14 (Category **Stoch**). Objects are measurable spaces. Morphisms are Markov kernels (Definition B.13). Composition is *Chapman–Kolmogorov* integration:

$$(\kappa * \lambda)(C \mid x) = \int_Y \kappa(C \mid y) \lambda(dy \mid x).$$

Definition B.15 (Giry Monad). Define $G : \mathbf{Meas} \rightarrow \mathbf{Meas}$ by $G(X, \mathcal{F}) = (PX, \mathcal{PF})$, where $\mathcal{PF}$ is the weak σ -algebra generated by $\mu \mapsto \mu(B)$ ($B \in \mathcal{F}$). For $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ set $Gf = f_*$

310 (Definition B.5). The *unit* $\eta_X : X \rightarrow GX$ is $x \mapsto \delta_x$ and the *multiplication* $\mu_X : GGX \rightarrow GX$
 312 is $\mu_X(\Phi)(B) = \int_{PX} \nu(B) d\Phi(\nu)$. With these, (G, η, μ) is a monad on **Meas** (Definition B.11); its
 Kleisli category is **Stoch** (Definition B.14).

Definition B.16 (Stochastic Dynamical System). A *stochastic dynamical system* is a G -coalgebra:
 314 a measurable map $c : X \rightarrow GX$ (Definition B.15). Iterating c via Kleisli composition yields a family
 of kernels $c^{(n)} : X \rightsquigarrow X$; see Definition B.17.

316 B.3 Markov Kernels & Stochastic Processes

Definition B.17 (Transition Kernel Family). A family $\{\kappa_t\}_{t \geq 0}$ of kernels $\kappa_t : X \rightsquigarrow X$ is a *transition*
 318 *kernel family* if

$$\kappa_{t+s} = \kappa_t * \kappa_s, \quad \kappa_0(x, \cdot) = \delta_x.$$

Definition B.18 (Discrete-Time Markov Chain). A *discrete-time Markov chain* on state space X
 320 is a pair (κ, μ_0) where $\kappa : X \rightsquigarrow X$ is a kernel and $\mu_0 \in PX$ an initial distribution. The law of X_{n+1}
 given $X_n = x$ is $\kappa(\cdot | x)$.

322 B.4 Semigroup & Generator Theory

Definition B.19 (Bounded Linear Operators). For a Banach space B , the set $\mathcal{L}(B)$ consists of all
 324 linear maps $A : B \rightarrow B$ with finite operator norm $\|A\|_{\mathcal{L}(B)} := \sup_{f \neq 0} \frac{\|Af\|_B}{\|f\|_B}$

Equipped with this norm, is itself a Banach space.

326 **Definition B.20** (Operator Semigroup). Let B be a Banach space. A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is
 called an *operator semigroup* (on B) if

$$P_0 = \text{id}_B, \quad P_{t+s} = P_t P_s \quad \text{for all } s, t \geq 0.$$

328 **Definition B.21** (Uniformly Continuous Semigroup). A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is a *uniformly*
continuous semigroup if

$$\lim_{t \downarrow 0} \|P_t - \text{id}\|_{\mathcal{L}(B)} = 0, \quad P_0 = \text{id}, \quad \text{for all } t \geq 0$$

330 Equivalently, $P_t = e^{tA}$ for a *bounded* operator $A \in \mathcal{L}(B)$ called the (bounded) generator, and the
 map $t \mapsto P_t$ is continuous in operator norm.

332 **Definition B.22** (Strongly Continuous Semigroup). A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is a *strongly con-*
tinuous semigroup (C_0 -semigroup) if

$$\lim_{t \downarrow 0} \|P_t f - f\|_B = 0, \quad \text{for all } f \in B, t \geq 0$$

334 **Definition B.23** (Markov Semigroup). A C_0 -semigroup $\{P_t\}_{t \geq 0}$ on $L^p(X, \lambda)$ is a *Markov semi-*
group if each P_t is positive and $P_t 1 = 1$.

336 **Definition B.24** (Infinitesimal Generator). For a C_0 -semigroup $\{P_t\}_{t \geq 0}$ the *generator* is the
(possibly unbounded) operator

$$Lf = \lim_{t \downarrow 0} \frac{P_t f - f}{t},$$

338 with domain $D(L) = \{f : \text{limit exists in } \|\cdot\|\}$.

C Locality in Infinitesimal Generators

340 C.1 The Evolution Kernel

In the classical theory of C_0 semigroups, the evolution operator $T(h)$ composes a C_0 semigroup
342 generated by a linear operator A

$$T(h) = e^{hA}.$$

In this setting, the kernel $K_h(\xi, \eta)$ is the Dirac delta

$$K_h(\xi, \eta) \xrightarrow{h \rightarrow 0} \delta(\xi - \eta).$$

344 *Proof.* Consider the integral operator

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta \quad (30)$$

with infinitesimal generator

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} (T(h) - I) f(\xi) \quad (31)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} ((T(h)f)(\xi) - f(\xi)) \quad (32)$$

By substitution

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta - f(\xi) \right) \quad (33)$$

The goal is to rewrite the integral term so that $f(\xi)$ appears inside the integral. We add and subtract $K_h(\xi, \eta) f(\xi)$ inside the integral,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) - K_h(\xi, \eta) f(\xi) + K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (34)$$

and factor,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \int_{\Omega} K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (35)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) \quad (36)$$

346 We can ensure that $K_h(\xi, \eta)$ acts as a state transition probability density by imposing the normalization condition,

$$\int_{\Omega} K_h(\xi, \eta) d\eta = 1 \quad (37)$$

348 while at the same time eliminating the second term,

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) = 0 \quad (38)$$

thus recovering the infinitesimal generator as a standard non-local operator,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta \quad (39)$$

350 **Kernel Structure:** The resolvent operator associated with A is given by:

$$R_{\lambda}(A) = (\lambda I - A)^{-1}. \quad (40)$$

It is well known (see, e.g. Pazy) that the semigroup $T(h)$ can be expressed in terms of the resolvent via the contour integral,

$$T(h) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_{\lambda}(A) d\lambda \quad (41)$$

352 We equate the integral representations

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_{\lambda}(A) f(\xi) d\lambda \quad (42)$$

Furthermore, Arendt & Bukhvalov showed that if $T(h)$ is an integral operator, then so is the resolvent,

$$(R_{\lambda}f)(\xi) = \int_{\Omega} M_{\lambda}(\xi, \eta) f(\eta) d\eta \quad (43)$$

354 where $M_{\lambda}(\xi, \eta)$ is the resolvent kernel. From (42) using the integral resolvent,

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \int_{\Omega} M_{\lambda}(\xi, \eta) f(\eta) d\eta d\lambda \quad (44)$$

Two cases follow regarding the absolute integrability of

$$F(\lambda, \eta) = e^{h\lambda} M_{\lambda}(\xi, \eta) f(\eta)$$

356 If $F \in L^1(\Gamma \times \Omega)$ then for

$$\oint_{\Gamma} \int_{\Omega} |F(\lambda, \eta)| d\mu(\eta) d\nu(\lambda) \quad (45)$$

we may apply Fubini's theorem and interchange the order of integration

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \int_{\Omega} \left(\frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_{\lambda}(\xi, \eta) d\lambda \right) f(\eta) d\eta \quad (46)$$

358 Since this must hold for all f , we identify the kernels,

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_{\lambda}(\xi, \eta) d\lambda \quad (47)$$

Since $T(0) = I$, the kernel at $h = 0$ must satisfy

$$K_0(\xi, \eta) = \delta(\xi - \eta) \quad (48)$$

360 In many classical settings, one can show that the resolvent satisfies

$$(\lambda I - A)R_{\lambda}(A) = I \quad (49)$$

so that in kernel form

$$(\lambda I - A)M_{\lambda}(\xi, \eta) = \delta(\xi - \eta) \quad (50)$$

362 $M_{\lambda}(\xi, \eta)$ serves as the Green's function for the operator $\lambda I - A$. Supposing that for values of λ in a suitable region the Neumann series expansion hold

$$(\lambda I - A)^{-1} = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \quad (51)$$

Then the kernel $M_{\lambda}(\xi, \eta)$ may be expressed as

$$(52)$$

$$M_{\lambda}(\xi, \eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \quad (53)$$

364 Substitute this expansion into

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \left(\sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \right) d\nu(\lambda) \quad (54)$$

$$= \delta(\xi - \eta) \sum_{n=0}^{\infty} A^n \left[\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) \right] \quad (55)$$

According to the residue theorem

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) = \frac{h^n}{n!} \quad (56)$$

366 Substituting this back into (55), we obtain

$$K_h(\xi, \eta) = \delta(\xi - \eta) \sum_{n=0}^{\infty} \frac{h^n}{n!} A^n = \delta(\xi - \eta) e^{hA} \quad (57)$$

Thus,

$$K_h(\xi, \eta) = \delta(\xi - \eta)e^{hA} \quad (58)$$

368 which is equivalent to the operator identity,

$$T(h) = e^{hA} \quad (59)$$

□

370 **The Nonlinear (State-Dependent) Case:** When the system is nonlinear, the evolution is typically given by a state-dependent mapping,

$$T(h)(\xi) = \Phi_h(\xi),$$

372 we may still maintain an integral representation:

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta.$$

In order for the kernel to capture the correct local behavior of the flow, we require that for small h ,
374

$$K_h(\xi, \eta) \rightarrow \delta(\Phi_h(\xi) - \Phi_h(\eta)).$$

Since Φ_0 is the identity mapping, this condition reduces to

$$\lim_{h \rightarrow 0} K_h(\xi, \eta) = \delta(\xi - \eta),$$

376 which, in the weak sense, means that for any test function f ,

$$\lim_{h \rightarrow 0} \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = f(\xi).$$

D A Categorical Scaffold for Local Linearization

378 Local linear models, residual covariances, and adaptive basis updates can all be written algebraically, but a categorical formulation clarifies *which parts belong to the dynamics, which to the coordinate*
380 *representation, and which to the data-driven closure*. By lifting the construction to functors, natural transformations, and (co)monads we obtain

- 382 (i) a precise separation of concerns
- (ii) a route to operator-theoretic results via functoriality
- 384 (iii) and a universal property (*cofree* coalgebra) that characterises the residual-frame augmentation

D.1 Components

Throughout, $k \geq 2$ is fixed, so that C^k flows have well-defined first and second jets. All manifolds are assumed compact, and all vector spaces are over $\mathbb{R}$.

Definition D.1 (Dyn). The category **Dyn** consists of

- **objects** $(\mathcal{M}, \varphi^t)$ where $\mathcal{M}$ is a compact C^k manifold and $\varphi^t : \mathcal{M} \rightarrow \mathcal{M}$ a C^k flow.
- **morphisms** $\Psi : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t)$ given by C^k maps $\Psi : \mathcal{M} \rightarrow \mathcal{N}$ satisfying

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \quad (60)$$

for all $t \in \mathbb{R}$.

Remark D.1. Objects in **Dyn** are smooth dynamical systems. Morphisms Ψ are *dynamic-preserving* (equivalently, *equivariant* or *trajectory-intertwining*). They are dynamic-preserving in the sense that for smooth manifolds and flows:

$$(\mathcal{M}, \varphi^t), (\mathcal{N}, \psi^t) \quad t \in \mathbb{R} \quad (61)$$

the morphism,

$$\Psi : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t) \quad (62)$$

is a C^k ($k \geq 1$) map that is equivariant,

$$\Psi(\varphi^t(x)) = \psi^t(\Psi(x)) \quad \forall x \in \mathcal{M} \quad (63)$$

so that Ψ transports trajectories φ^t onto trajectories ψ^t :

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \quad (64)$$

Example D.1. Take the two-torus with linear flow,

$$(\mathbb{T}^2, \varphi^t), \quad \varphi^t(x, y) = (x + \alpha t, y + \beta t) \pmod{1} \quad (65)$$

and the circle with rigid rotation,

$$(\mathbb{S}^1, \psi^t), \quad \psi^t(\theta) = \theta + \alpha t \pmod{1} \quad (66)$$

Let,

$$\Psi : \mathbb{T}^2 \rightarrow \mathbb{S}^1, \quad \Psi(x, y) = x \pmod{1} \quad (67)$$

For all $(x, y) \in \mathbb{T}^2$ $t \in \mathbb{R}$:

$$\Psi(\varphi^t(x, y)) = \Psi(x + \alpha t, y + \beta t) \quad (68)$$

$$= (x + \alpha t) \pmod{1} \quad (69)$$

$$= \psi^t(\Psi(x, y)) \quad (70)$$

so,

$$\Psi \circ \varphi^t = \psi^t \circ \Psi, \quad \forall t \in \mathbb{R} \quad (71)$$

and Ψ is a dynamic-preserving C^∞ map and morphism of **Dyn**.

404 *Remark D.2* (Discrete-time systems via suspension). Although **Dyn** is defined only for *flows*, any discrete-time diffeomorphism $T: \mathcal{M} \rightarrow \mathcal{M}$ can be embedded in **Dyn** through its *suspension flow*.

406 Form the mapping torus

$$\text{Sus}(\mathcal{M}, T) := (\mathcal{M} \times [0, 1]) / (x, 1) \sim (Tx, 0), \quad (72)$$

and define the flow

$$\varphi^t[(x, u)] = [(T^{\lfloor t+u \rfloor} x, u + t - \lfloor t+u \rfloor)] \quad (73)$$

408 Then $(\text{Sus}(\mathcal{M}, T), \varphi^t)$ is an object of **Dyn** whose *time-1 map* reproduces T :

$$\varphi^1[(x, 0)] = [(Tx, 0)] \quad (74)$$

This construction extends to morphisms by $\Sigma(f)[(x, u)] = [(f(x), u)]$, producing a functor $\Sigma: \mathbf{Dyn}_{\mathbb{Z}} \rightarrow \mathbf{Dyn}$. As such, results for continuous-time objects apply *mutatis mutandis* to discrete systems via Σ .

412 **Definition D.2** (**Vect**). The category **Vect** consists of

- **objects** V finite-dimensional real vector spaces.
- **morphisms** $L: V \rightarrow W$ linear maps.

414 **Definition D.3** (Embedding functor). Fix, for every **Dyn**-object $(\mathcal{M}, \varphi^t)$ a generic embedding $\Phi_{\mathcal{M}}: \mathcal{M} \hookrightarrow \mathbb{R}^{m(\mathcal{M})}$. Define the functor

$$F: \mathbf{Dyn} \longrightarrow \mathbf{Vect} \quad \text{by} \quad \begin{cases} F(\mathcal{M}, \varphi^t) = \mathbb{R}^{m(\mathcal{M})}, \\ F(\Psi) = \Phi_{\mathcal{N}} \circ \Psi \circ \Phi_{\mathcal{M}}^{-1}: \mathbb{R}^{m(\mathcal{M})} \longrightarrow \mathbb{R}^{m(\mathcal{N})}. \end{cases}$$

with identity $F(\text{id}_{(\mathcal{M}, \varphi^t)}) = \text{id}_{\mathbb{R}^{m(\mathcal{M})}}$.

418 Functoriality is immediate from $\Phi_{\mathcal{N}} \circ (\Psi_2 \circ \Psi_1) = F(\Psi_2) \circ F(\Psi_1)$, which gives the following diagram:

$$\begin{array}{ccc} (\mathcal{M}, \varphi^t) & \xrightarrow{\Phi := \Phi_{\mathcal{M}}} & \mathbb{R}^{m(\mathcal{M})} \\ \Psi \downarrow & & \downarrow \Phi' \circ \Psi \circ \Phi^{-1} \\ (\mathcal{N}, \psi^t) & \xrightarrow{\Phi' := \Phi_{\mathcal{N}}} & \mathbb{R}^{m(\mathcal{N})} \end{array}$$

Definition D.4 (**Dyn** Component-wise product). Let $(\mathcal{M}_i, \varphi_i^t)_{i=1}^n$ be objects of **Dyn**. Define

$$\mathcal{M} := \prod_{i=1}^n \mathcal{M}_i, \quad \varphi^t(x_1, \dots, x_n) := (\varphi_1^t(x_1), \dots, \varphi_n^t(x_n)), \quad (75)$$

420 and projection maps $\pi_j(x_1, \dots, x_n) = x_j$ ($j = 1, \dots, n$). We call

$$(\mathcal{M}, \varphi^t) \quad (76)$$

the *component-wise product* of the given objects.

422 **Lemma D.1** (Universal property of finite products in **Dyn**). *For the system $(\mathcal{M}, \varphi^t)$ and projections π_j defined above, the pair*

$$((\mathcal{M}, \varphi^t), \{\pi_j\}_{j=1}^n) \quad (77)$$

424 *is the categorical product of $(\mathcal{M}_i, \varphi_i^t)_{i=1}^n$ in the category **Dyn**.*

Proof. Because each π_j is smooth,

$$\pi_j(\varphi^t(x_1, \dots, x_n)) = \varphi_j^t(x_j) = \varphi_j^t(\pi_j(x_1, \dots, x_n)), \quad (78)$$

426 so π_j intertwines the flows and is therefore a morphism in **Dyn**.

Let $(\mathcal{N}, \psi^t)$ be any object and suppose we are given morphisms $f_j : (\mathcal{N}, \psi^t) \rightarrow (\mathcal{M}_j, \varphi_j^t)$ for $j = 1, \dots, n$. Define

$$f : \mathcal{N} \longrightarrow \mathcal{M}, \quad f(y) := (f_1(y), \dots, f_n(y)). \quad (79)$$

The map f is C^k because each f_j is. Equivariance follows component-wise:

$$f(\psi^t y) = (f_1(\psi^t y), \dots) = (\varphi_1^t f_1(y), \dots) = \varphi^t(f(y)). \quad (80)$$

430 Hence f is a morphism in **Dyn** and $\pi_j \circ f = f_j$ by construction.

If $g : (\mathcal{N}, \psi^t) \rightarrow (\mathcal{M}, \varphi^t)$ is another morphism with the same property $\pi_j \circ g = f_j$ for every j , then for each $y \in \mathcal{N}$

$$g(y) = (\pi_1 g(y), \dots, \pi_n g(y)) = (f_1(y), \dots, f_n(y)) = f(y),$$

so $g = f$.

434 Since a unique mediating morphism exists for every candidate cone, $(\mathcal{M}, \varphi^t)$ with the projections π_j satisfies the universal property of finite products in **Dyn**. $\square$

436 **Lemma D.2** (Coproducts). *For any family $(\mathcal{M}_i, \varphi_i^t)_{i \in I}$, the disjoint union*

$$\left(\bigsqcup_{i \in I} \mathcal{M}_i, \varphi^t(x) = \varphi_i^t(x) \text{ if } x \in \mathcal{M}_i \right)$$

*with the canonical injections $\iota_i : \mathcal{M}_i \hookrightarrow \bigsqcup_j \mathcal{M}_j$ is the coproduct in **Dyn**.*

438 *Proof.* Disjoint unions of compact manifolds are compact manifolds, the flow is C^k on each component, and the injections are equivariant by construction. The universal property follows from the universal property of disjoint unions in smooth manifolds. $\square$

442 **Lemma D.3** (Regular equalisers). *Let $f, g : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t)$ be morphisms. Assume f and g are transverse to the diagonal $\Delta_{\mathcal{N}} \subset \mathcal{N} \times \mathcal{N}$. Then*

$$E := \{x \in \mathcal{M} \mid f(x) = g(x)\}$$

444 *is an embedded C^k sub-manifold, invariant under φ^t . The inclusion $i : E \hookrightarrow \mathcal{M}$ with the restricted flow $\varphi|_E^t$ realises the equaliser of (f, g) in **Dyn**.*

Proof. Transversality implies E is an embedded sub-manifold. Equivariance of f and g gives
 446 $f(\varphi^t x) = \psi^t f(x) = \psi^t g(x) = g(\varphi^t x)$, so E is flow-invariant. The inclusion i is a morphism,
 448 and any morphism $h : (\mathcal{K}, \chi^t) \rightarrow (\mathcal{M}, \varphi^t)$ with $fh = gh$ factors uniquely through i by the usual
 smooth-manifold argument. $\square$

Remark D.3 (Limitations of completeness). The category of C^k manifolds is neither complete nor
 450 cocomplete; an infinite product need not be a manifold, and general coequalisers (or non-regular
 equalisers) can fail to carry a smooth Hausdorff structure. Consequently **Dyn** is *finitely* complete
 452 and has coproducts, but it does not possess arbitrary small limits or colimits.

Lemma D.4 (Limits and colimits). **Dyn** has all small limits and colimits; they are inherited
 454 point-wise from the category of C^k manifolds.

Sketch. Products exist because finite products of compact manifolds are compact manifolds; the
 456 flow acts component-wise. Equalisers are submanifolds defined by equalising maps that commute
 with the flow. Dually for colimits. $\square$

458 The remainder of this section builds functors *out of Dyn* and uses natural transformations and
 comonads to formalise local linearisation and its residual correction.

460 D.2 Local linearisation as a natural transformation

Write $\Pi_d : \mathbf{Dyn} \rightarrow \mathbf{Dyn}$ for the functor that forgets all ≥ 2 -jet data. Weighted least squares at
 462 each anchor x induces the component

$$\eta_{(\mathcal{M}, \varphi^t)}(x) := B_x : F(\mathcal{M}, \varphi^t) \longrightarrow F \circ \Pi_d(\mathcal{M}, \varphi^t),$$

yielding a natural transformation $\eta : F \Rightarrow F \circ \Pi_d$.

464 D.3 Residual frame field as a cofree coalgebra

Define a comonad $\mathbf{G} : (\mathbf{Vect}, \otimes) \rightarrow (\mathbf{Vect}, \otimes)$ by $\mathbf{G}(V) = V \oplus (\text{span } Q_x)$. The map $x \mapsto (B_x, Q_x)$
 466 endows each fibre with the structure of a $\mathbf{G}$ -coalgebra, cofree under the variance-fraction constraint
 ρ . In string-diagram notation:

$$\begin{array}{ccc} F(\mathcal{M}, \varphi^t) & \xrightarrow{\quad \Delta \quad} & \mathbf{G}F(\mathcal{M}, \varphi^t) \\ \eta \downarrow & & \downarrow \mathbf{G}\eta \\ F \circ \Pi_d(\mathcal{M}, \varphi^t) & \xrightarrow{\quad \mathbf{G}\Delta \quad} & \mathbf{G}F \circ \Pi_d(\mathcal{M}, \varphi^t) \end{array}$$

468 The dashed arrows are the Rose augmentation; commutativity is enforced by letting $\mathbf{G}$ be *cofree*
 with respect to ρ .